Multiple Linear Regression Predictive Model

Verified testing, model metrics, and June 2026 forecast summary

Method and Test Split

This report presents a Multiple Linear Regression model for predicting air temperature and PM10 concentration. The verified test period uses the latest 15 days of the available hourly modelling dataset, from 2026-04-16 00:00:00 to 2026-05-01 00:00:00. Earlier rows are used for training.

For the verified test, each target is predicted using the other measured columns, time features, wind direction sine/cosine features, and lag/rolling features. The current target column itself is excluded from its own feature list. The June forecast is not scored because June measured values were not provided in the modelling dataset.

Verified Test Metrics

Target	MAE	MSE	RMSE	R2	Mean Rel. Error
Temperature	0.4068	0.2902	0.5387	0.9861	2.78%
PM10	1.6246	5.8981	2.4286	0.7498	11.18%

The model reached MAE 0.4068 C and RMSE 0.5387 C for temperature. For PM10 it reached MAE 1.6246 ug/m3 and RMSE 2.4286 ug/m3. Lower MAE, MSE, RMSE, and mean relative error are better. R2 is better when it is closer to 1.

Metric Definitions

MAE is the average absolute error. MSE is the average squared error and gives extra penalty to large mistakes. RMSE is the square root of MSE, so it is easier to interpret in the original unit. R2 measures how much variation is explained by the model.

Dataset Range

2023-03-17 11:00:00
to 2026-05-01 00:00:00

Source Rows

24096

Held-Out Test

361 rows
latest 15 days

June Forecast

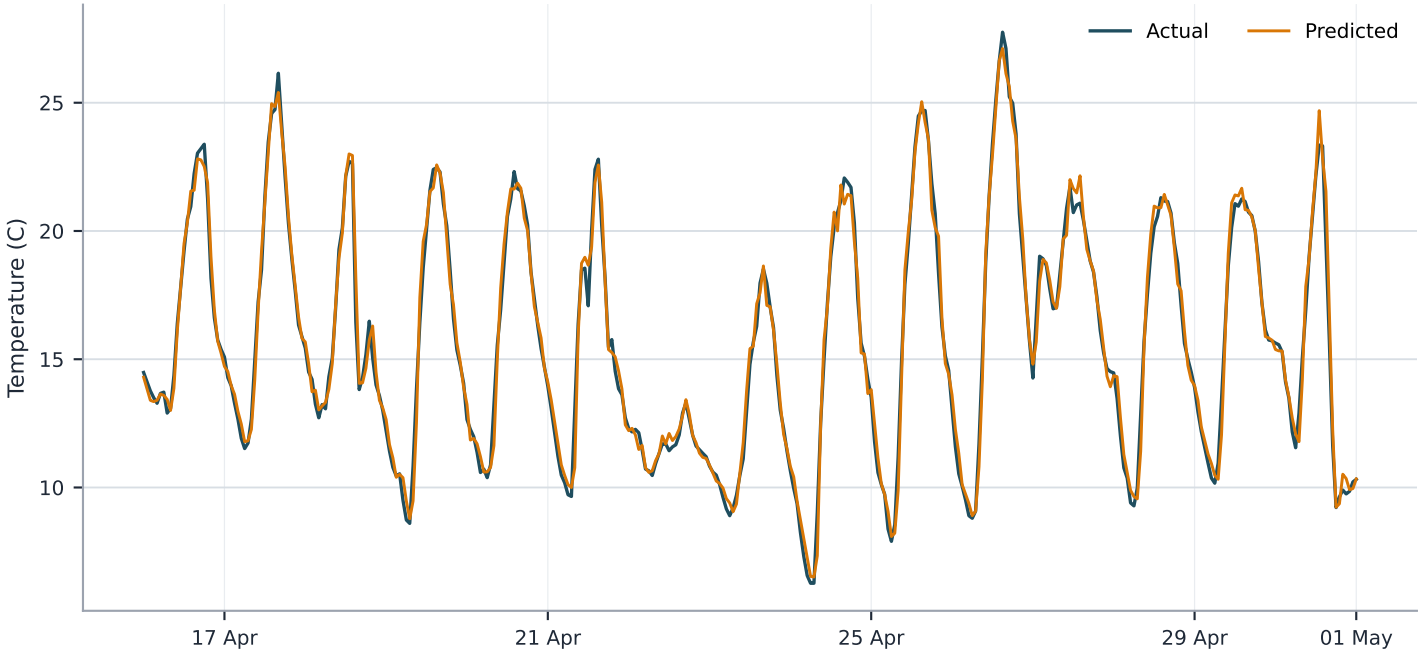
2026-06-01 00:00
to 2026-06-30 23:00

Verified Test: Actual vs Predicted

The same held-out timestamps are used for both targets.

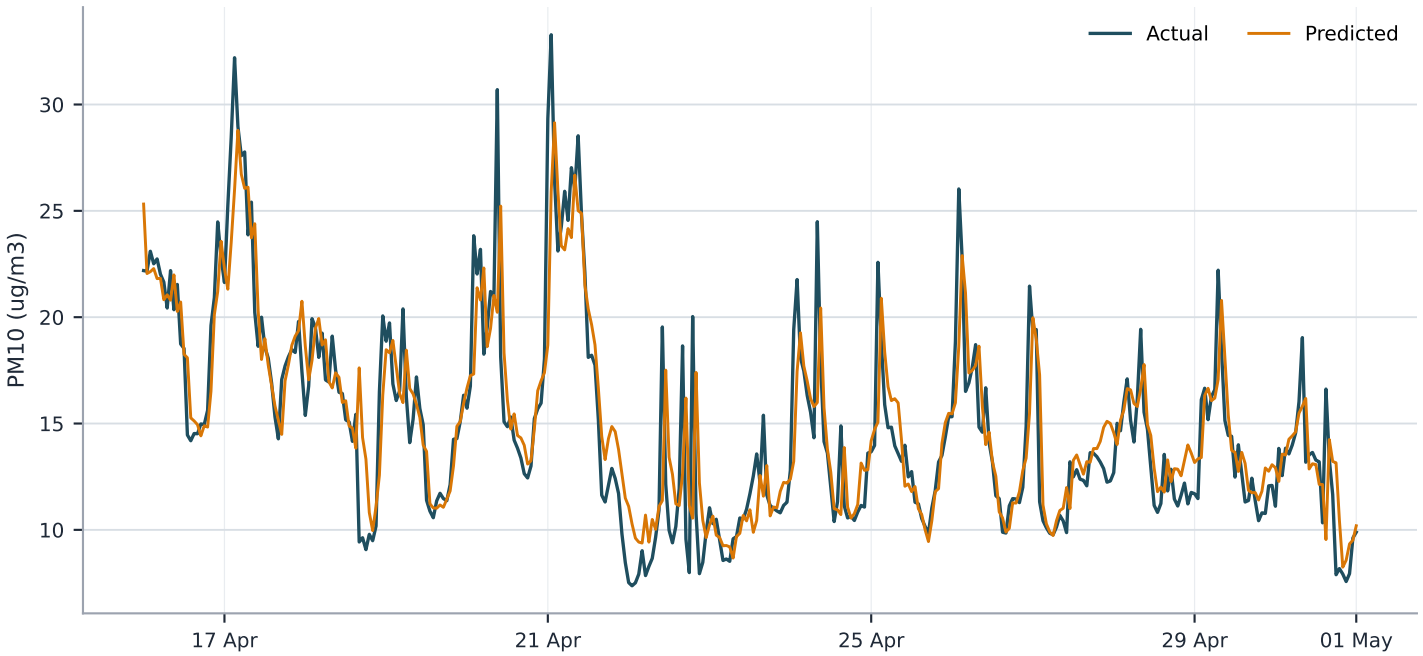
Temperature

Actual and predicted air temperature during the verified test period.



PM10

Actual and predicted PM10 concentration during the verified test period.

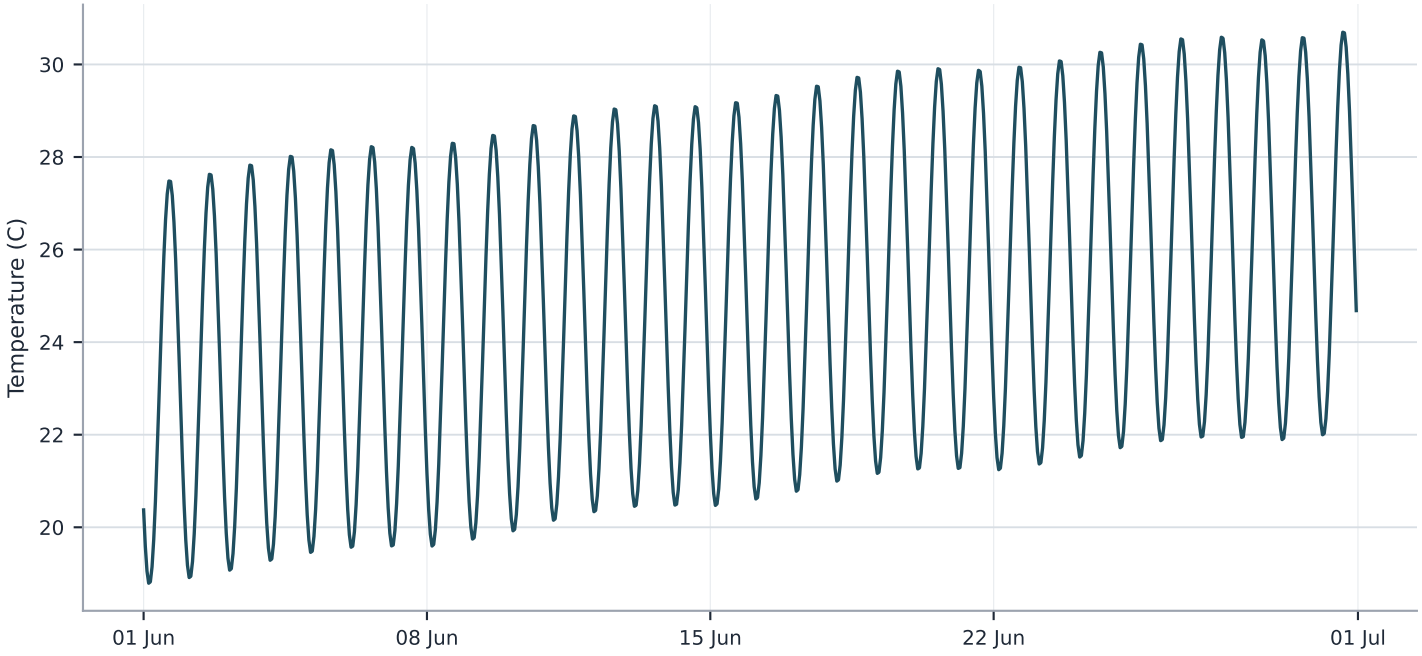


June 2026 Forecast

Forecast-only output from 2026-06-01 00:00 to 2026-06-30 23:00.

Predicted Temperature

Hourly forecast produced by the Multiple Linear Regression model.



Predicted PM10

Hourly forecast produced by the Multiple Linear Regression model.

